

Design a Web Crawler - Extensibility

1. What is Extensibility?

- Extensibility means the system can:
 - **Evolve over time**
 - **Support new features or content types**

Without redesigning the entire system

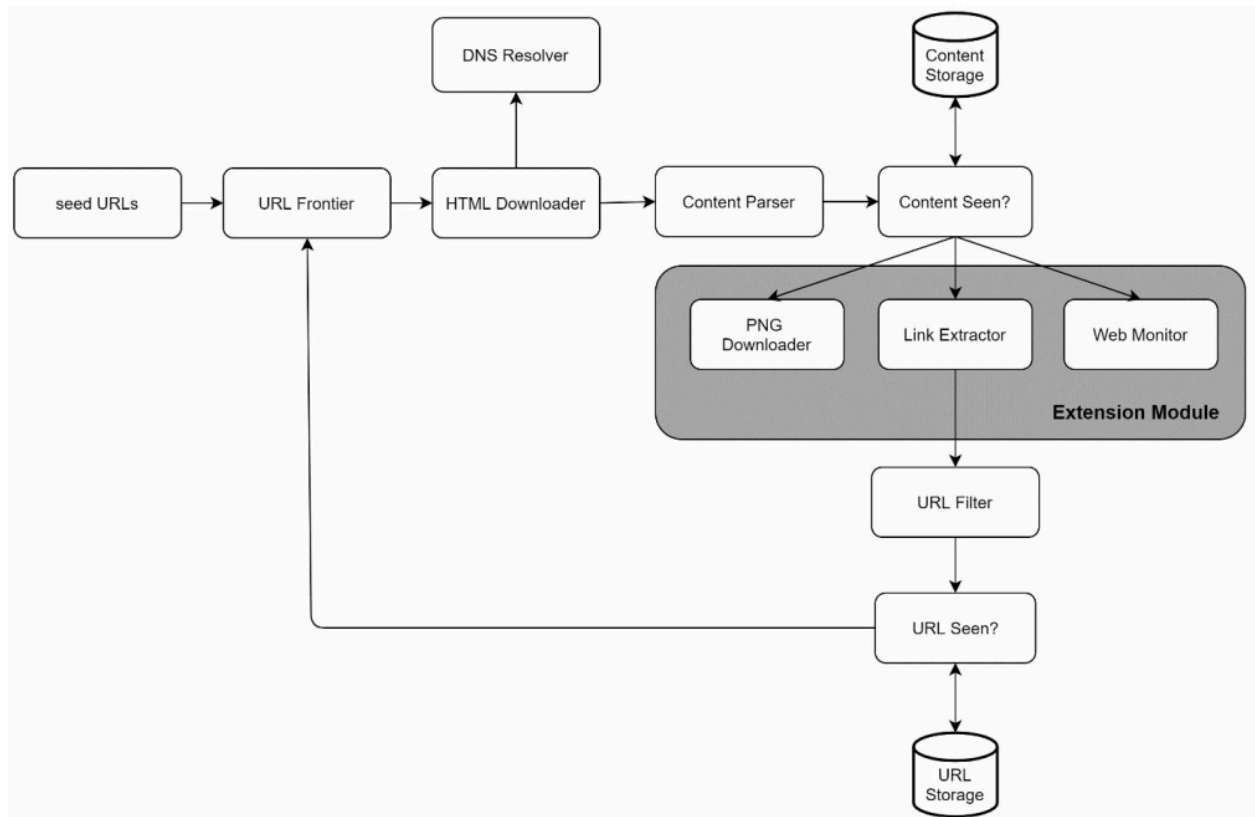
2. Design Goal

- Make crawler **flexible and modular**

Achieved by:

- Adding new functionality via **plug-in modules**

3. Modular Architecture



- System is designed in a way that:
 - New modules can be **plugged in easily**
 - Existing components remain unchanged

4. Example Extensions

4.1 PNG Downloader Module

Purpose

- Extend crawler to support:

- **PNG image files**

Function

- Downloads image content instead of HTML

Benefit

- Supports **new content type (images)**
- No need to redesign crawler

4.2 Web Monitor Module

Purpose

- Monitor web content for:
 - Copyright violations
 - Trademark infringements

Function

- Scans crawled data
- Detects misuse of protected content

Benefit

- Adds **monitoring capability**
- Useful for legal and compliance use cases

5. Key Takeaway

Extensibility allows:

- Easy addition of new features
- Support for new content types
- System evolution without major redesign

Achieved through:

- Modular design
- Plug-in architecture